CONDITIONS	RULES		
Is the minimum interval date ≤ date administered?	No	No	Yes
OUTCOMES	No. The vaccine dose administered did not satisfy the preferable interval for the target dose. Evaluation reason is 'Too soon'.	Yes. The vaccine dose administered satisfied the preferable interval for the target dose. Evaluation reason is 'Grace period'.	Yes. The vaccine dose administered satisfied the preferable interval for the target dose.

Table 6-19 Evaluate Preferable Interval Business Rules

Business Rule ID	Business Rule
CALCDTINT-1	A patient's reference dose date for an interval must be calculated as the date administered of the most immediate previous vaccine dose administered if all the following are true: • the interval has a from immediate previous dose administered flag of 'Y' • the vaccine dose administered has an evaluation status of 'Valid' or 'Not Valid' • the vaccine dose administered is not an inadvertent administration.
CALCDTINT-2	A patient's reference dose date for an interval must be calculated as the date administered of the vaccine dose administered that satisfies the target dose with the same target dose number as the from target dose number in series if all the following are true for the interval: • from immediate previous dose administered flag is 'N' • from target dose number in series is not 'N/A'.
CALCDTINT-3	A patient's absolute minimum interval date must be calculated as the patient's reference dose date plus the absolute minimum interval.
CALCDTINT-4	A patient's minimum interval date must be calculated as the patient's reference dose date plus the minimum interval.
CALCDTINT-8	A patient's reference dose date for an interval must be calculated as the date administered of the most recent vaccine dose administered that is the same vaccine type as the from most recent vaccine type if all the following are true for the interval: • from immediate previous dose administered flag is 'N' • from most recent vaccine type is not 'N/A' • the vaccine dose administered is not an inadvertent administration.
CALCDTINT-9	A patient's reference dose date for an interval must be calculated as the observation date of the most recent active patient observation if all the following are true for the interval: • from immediate previous dose administered flag is 'N' • from relevant observation code is not 'N/A'.

6.6 Evaluate Allowable Interval

Evaluate allowable interval validates the date administered of a vaccine dose administered against defined allowable interval(s) from previous vaccine dose(s) administered. In rare cases, intervals can be applied which are either abnormally early – usually specified in ACIP footnotes or subsequent clarifications – or intervals which differ following a not valid administration.

In cases where a target dose does not specify allowable interval attributes, evaluate allowable interval cannot be used to validate a vaccine dose administered. To avoid a false validation, the allowable interval should be considered “not valid” in these cases.

The figure below provides evaluate allowable interval timeline used to define all adjacent intervals by using *from immediate previous dose administered flag* as the reference dose.

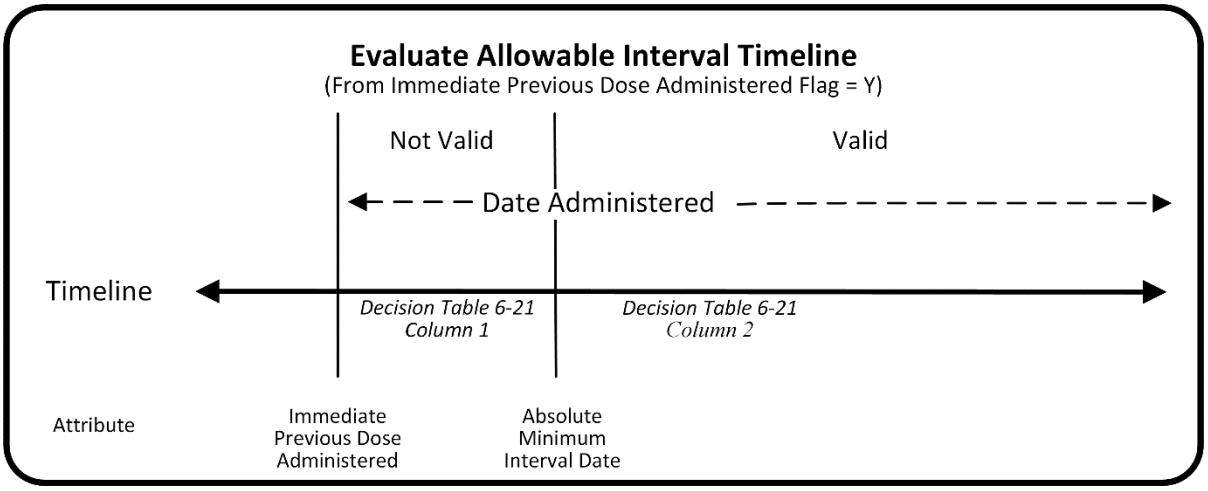

FIGURE 6-12 EVALUATE ALLOWABLE INTERVAL 'FROM IMMEDIATE PREVIOUS DOSE ADMINISTERED FLAG' TIMELINE

The figure below illustrates evaluate allowable interval timeline used to define all non-adjacent intervals by using *from target dose number in series* as the reference dose.

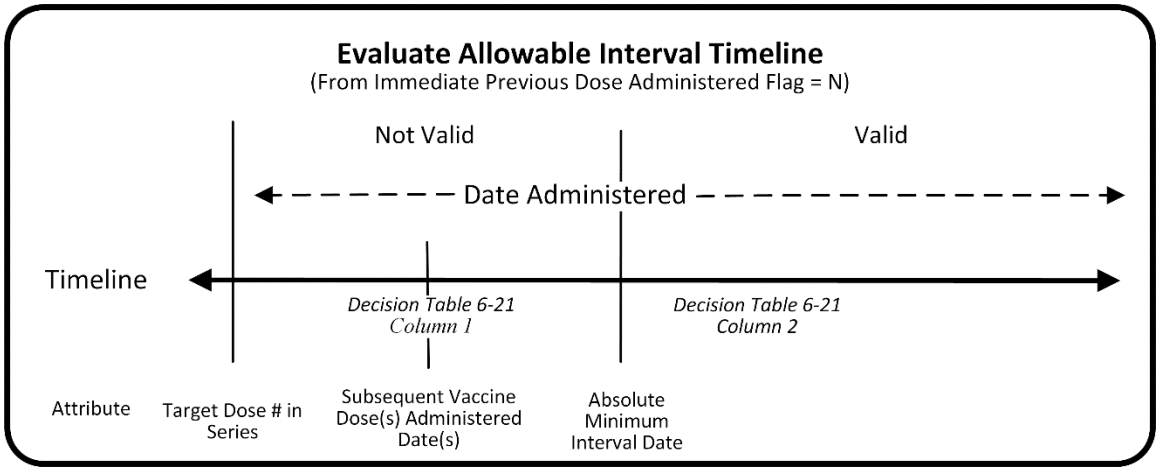

FIGURE 6-13 EVALUATE ALLOWABLE INTERVAL 'FROM TARGET DOSE NUMBER IN SERIES' TIMELINE

The following process model, attribute table, decision table, and business rule table are used to evaluate interval of a vaccine dose administered.

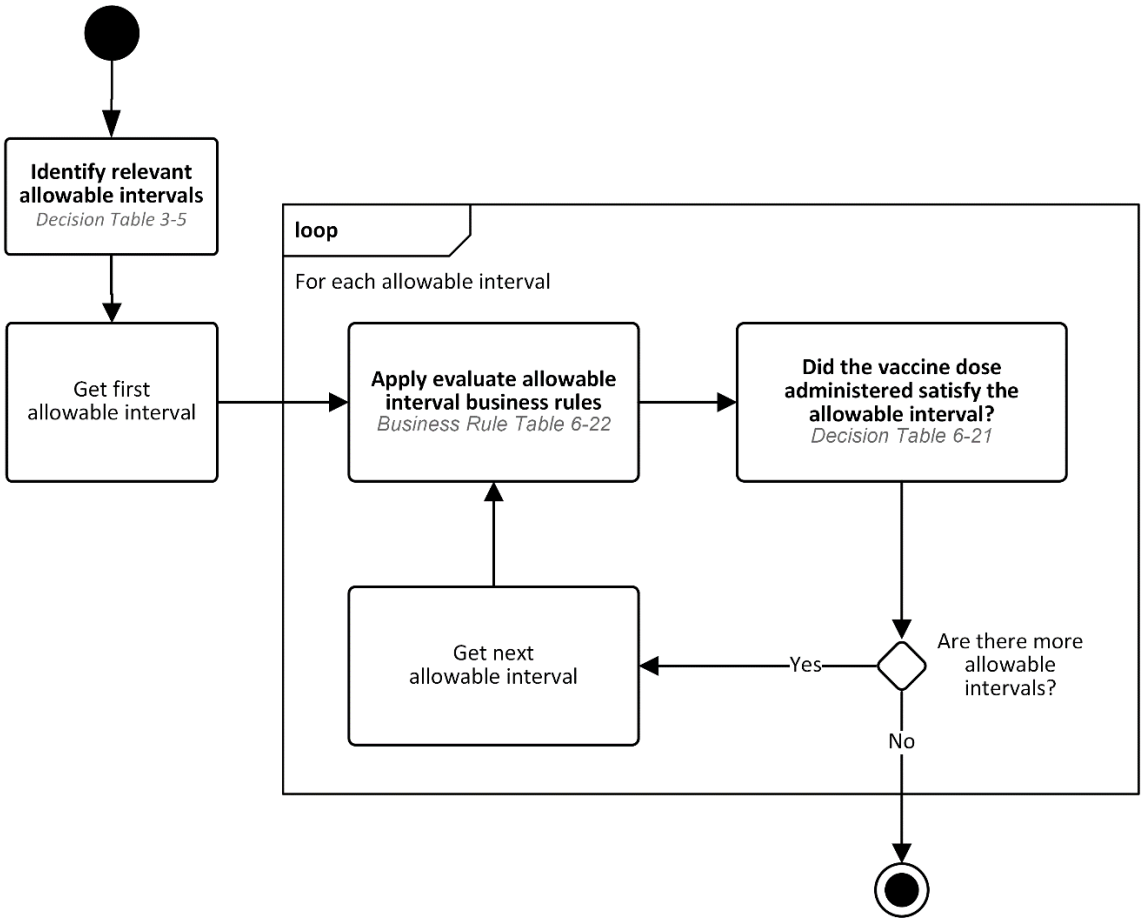

FIGURE 6-14 EVALUATE ALLOWABLE INTERVAL PROCESS MODEL

TABLE 6-20 ALLOWABLE INTERVAL ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Vaccine dose administered	Date Administered	-
Supporting Data	Allowable Interval elements	-
Calculated date (CALCDTINT-3)	Absolute Minimum Interval Date	01/01/1900

TABLE 6-21 DID THE VACCINE DOSE ADMINISTERED SATISFY THE ALLOWABLE INTERVAL FOR THE TARGET DOSE?

CONDITIONS	RULES	
Is the date administered < absolute minimum interval date?	Yes	No
OUTCOMES	No. The vaccine dose administered did not satisfy the allowable interval for the target dose. Evaluation Reason is 'Too soon'.	Yes. The vaccine dose administered satisfied the allowable interval for the target dose.

TABLE 6-22 EVALUATE ALLOWABLE INTERVAL BUSINESS RULES

Business Rule ID	Business Rule
CALCDTINT-1	A patient's reference dose date for an interval must be calculated as the date administered of the most immediate previous vaccine dose administered if all the following are true: • The interval has a from immediate previous dose administered flag of 'Y' • The vaccine dose administered has an evaluation status of 'Valid' or 'Not Valid' • The vaccine dose administered is not an inadvertent administration.
CALCDTINT-2	A patient's reference dose date for an interval must be calculated as the date administered of the vaccine dose administered that satisfies the target dose with the same target dose number as the from target dose number in series if all the following are true for the interval: • From immediate previous dose administered flag is 'N' • From target dose number in series is not 'n/a'
CALCDTINT-3	A patient's absolute minimum interval date must be calculated as the patient's reference dose date plus the absolute minimum interval.

6.7 Evaluate Vaccine Conflict

Evaluate vaccine conflict validates the date administered of a vaccine dose administered against previous administered vaccines to ensure proper spacing between administrations. This covers live virus vaccine conflicts as well as non-live virus vaccine conflicts. Many vaccines do not have any conflict with each other. Therefore, if no vaccine Supporting Data exists for the vaccine type of the vaccine dose administered being evaluated, the vaccine dose administered is not in conflict with any other vaccine dose administered.

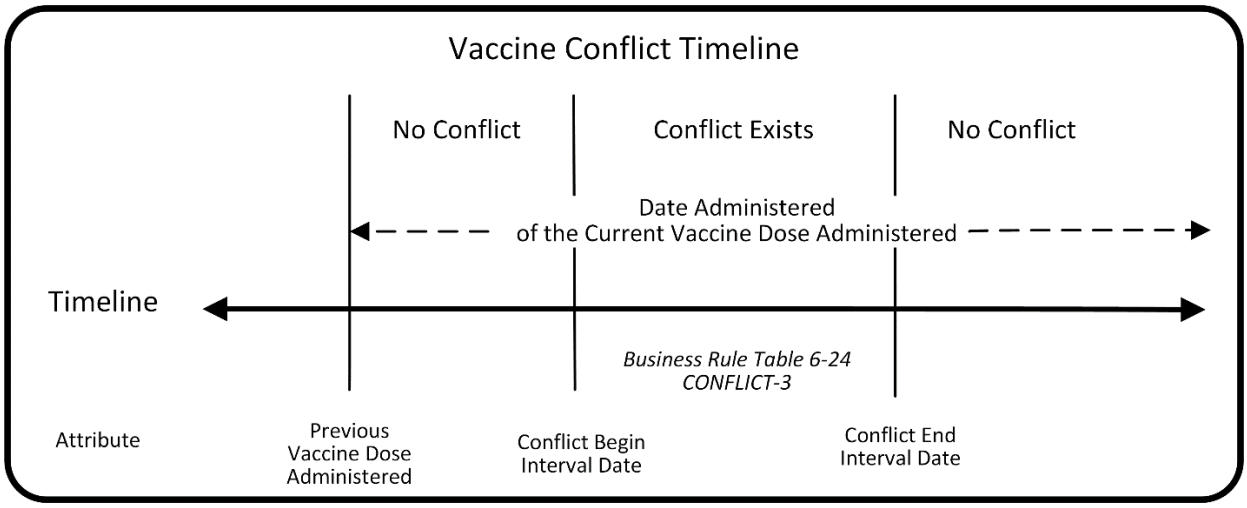

FIGURE 6-15 EVALUATE VACCINE CONFLICT TIMELINE

The following process model, attribute table, decision tables, and business rule table are used to evaluate for a vaccine conflict.